



UNIVERSITY OF AGRICULTURAL SCIENCES-560065
Department of Soil Science and Agricultural Chemistry, G.K.V.K.
BENGALURU

SS&AC/ /2025-26
Ref: DR/SO/TT- /2025-26 Dt.

ಮಣ್ಣು ಆರೋಗ್ಯ ಪತ್ರ

Dt. 26/02/26

Details/ ವಿವರ			
Soil Test report No.	408	Transaction ID	T2602261441150976759183
Name/ ಹೆಸರು	VEDHA BHOOMI FARM LAND		
Address/ವಿಳಾಸ	Vedashree Parivar LLP., 2 nd Floor, GO-PIZZA LINE, Unit No. 852, 8 th Main Road, Sahakar Nagara, Bangalore-560092		
MOBILE NO/ಮೊಬೈಲ್ ಸಂಖ್ಯೆ	9900014408/ 9591712266		
Sample Details/ಮಾದರಿಯ ವಿವರ	Soil		
Survey No. & Sample Collected Area	842, Revenue Village, Chilmattur Mandal, Sri Sathya Sai District, AP		
Gmail ID			

Soil Analysis Results/ಮಣ್ಣು ವಿಶ್ಲೇಷಣಾ ವರದಿ

Paramaters/ನಿಯತಾಂಕಗಳು		ಪ್ರಮಾಣ 1	ಷರಾ	
pH (1:2.5)	ರಸಸಾರ	6.84	Normal	Potentiometry Jackson (1973)
EC (dS/m)	ವಿದ್ಯುತ್ ವಾಹಕತ್ವ, ಡೆಸಿ ಸೈಮನ್/ಮೀ	0.15	Low/ಕಡಿಮೆ	Conductometry Jackson (1973)
Organic Carbon (%)	ಸಾವಯವ ಇಂಗಾಲ (%)	0.40	Low/ಕಡಿಮೆ	Wet digestion Walkley and black (1934)
Major Nutrients/ಪ್ರಧಾನ ಪೋಷಕಾಂಶಗಳು				
Available Nitrogen (Kg ha ⁻¹)	ದೊರೆಯುವ ಸಾರಜನಕ(N) ಕೆ.ಜಿ./ಹೆ.	188.15	Low/ಕಡಿಮೆ	Kjeldalh method Subbaih and Asija(1956)
Available Phosphorus (Kg ha ⁻¹)	ದೊರೆಯುವರಂಜಕ(P ₂ O ₅) ಕೆ.ಜಿ./ಹೆ.	18.44	Low/ಕಡಿಮೆ	Olsen method Brays and kurtz (1945)
Available Potassium (Kg ha ⁻¹)	ದೊರೆಯುವ ಪೊಟ್ಯಾಶ್(K ₂ O) ಕೆ.ಜಿ./ಹೆ.	155.00	Medium/ಮಧ್ಯಮ	Flame photometry Jackson (1973)
Secondary Nutrients/ದ್ವಿತೀಯ ಪೋಷಕಾಂಶಗಳು				
Exchangeable Calcium (Meq/100g)	ವಿನಿಮಯಕಾರ್ಕಲ್ಷಿಯಂ (ಮಿ.ಈ/100ಗ್ರಾಂ)	3.61	High/ಹೆಚ್ಚು	Titrimetry Barauh and Barthakur (1997)
Exchangeable Magnesium(Meq/100g)	ವಿನಿಮಯ ಮೆಗ್ನೀಷಿಯಂ (ಮಿ.ಈ/100ಗ್ರಾಂ)	1.50	Medium/ಮಧ್ಯಮ	Titrimetry Barauh and Barthakur (1997)
Available Sulphur (ppm)	ದೊರೆಯುವಗಂಧಕ (ಪಿ.ಪಿ.ಎಂ)	11.00	Medium/ಮಧ್ಯಮ	Turbidometry Black(1997)
Micro Nutrients/ಲಘು ಪೋಷಕಾಂಶಗಳು				
Iron (ppm)	ಕಬ್ಬಿಣ (ಪಿ.ಪಿ.ಎಂ)	6.51	High/ಹೆಚ್ಚು	DTPA , Lindsay and norwell(1978)
Manganese (ppm)	ಮ್ಯಾಂಗನೀಸ್ (ಪಿ.ಪಿ.ಎಂ)	3.44	High/ಹೆಚ್ಚು	DTPA , Lindsay and orwell(1978)
Copper (ppm)	ತಾಮ್ರ (ಪಿ.ಪಿ.ಎಂ)	2.18	High/ಹೆಚ್ಚು	DTPA , Lindsay and norwell(1978)
Zinc (ppm)	ಸತು (ಪಿ.ಪಿ.ಎಂ)	0.41	Low/ಕಡಿಮೆ	DTPA , Lindsay and norwell(1978)

- ಮಣ್ಣು ಮಾದರಿಯನ್ನು ರೈತರು ಸ್ವತಃ ಸಂಗ್ರಹಿಸಿ ಪ್ರಯೋಗಾಲಯಕ್ಕೆ ತಂದಿರುತ್ತಾರೆ.
- ಈ ವರದಿಯು ಯಾವುದೇ ಕಾನೂನು ಸಂಭಂದಿತ ವ್ಯವಹಾರಗಳಿಗೆ ಅನ್ವಯಿಸುವುದಿಲ್ಲ

Soil Analyst: Ramya Krishna K

ಪ್ರಾಧ್ಯಾಪಕರು ಮತ್ತು ಮುಖ್ಯಸ್ಥರು



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Sample Details/ಮಾದರಿಯ ವಿವರ	Water		
Survey No. & Sample Collected Area	842, Revenue Village, Chilmattur Mandal, Sri Sathya Sai District, AP		
GST No.			

➤ **Irrigation Water Analysis Results/ನೀರಾವರಿ ನೀರಿನವಿಶ್ಲೇಷಣಾ ವರದಿ**

Paramaters/ನಿಯತಾಂಕಗಳು		RESULTS ಪ್ರಮಾಣ	INTERPRETATION ಷರಾ
pH (1:2.5)	ರಸಸಾರ	7.01	Medium/ಮಧ್ಯಮ
EC (dS/m)	ವಿದ್ಯುತ್ ವಾಹಕತ್ವ, ಡೆಸಿ, ಸೈಮನ/ಮೀ	1.99	High/ ಹೆಚ್ಚು
Carbonates (CO ₃) (meq/L)	ಕಾರ್ಬೋನೇಟ್ಸ್ (ಮಿ.ಈ/ಲೀ)	-	-
Bicarbonates (CO ₃) (meq/L)	ಬೈಕಾರ್ಬೋನೇಟ್ಸ್ (ಮಿ.ಈ/ಲೀ)	7.70	-
Chlorides (Cl) (meq/L)	ಕ್ಲೋರೈಡ್ಸ್ (ಮಿ. ಗ್ರಾಂ/ಲೀ)	10.40	Medium/ಮಧ್ಯಮ
Calcium (Ca) (meq/L)	ಕ್ಯಾಲ್ಸಿಯಂ (ಮಿ.ಈ/ಲೀ)	9.00	-
Magnesium (Mg) (meq/L)	ಮೆಗ್ನೀಷಿಯಂ (ಮಿ.ಈ/ಲೀ)	7.40	-
Sodium (Na) (meq/L)	ಸೋಡಿಯಂ (ಮಿ.ಈ/ಲೀ)	2.77	-
SAR	ಸೋಡಿಯಂ ಹೊರಹೀರುವಿಕೆ ಅನುಪಾತ	0.96	Low/ಕಡಿಮೆ
RSC	ಉಳಿಕೆ ಸೋಡಿಯಂ ಕಾರ್ಬೋನೇಟ್	-	-

- ನೀರಿನ ಮಾದರಿಯನ್ನು ರೈತರು ಸ್ವತಃ ಸಂಗ್ರಹಿಸಿ ಪ್ರಯೋಗಾಲಯಕ್ಕೆ ತಂದಿರುತ್ತಾರೆ.
➤ ಈ ವರದಿಯು ಯಾವುದೇ ಕಾನೂನು ಸಂಭಂದಿತ ವ್ಯವಹಾರಗಳಿಗೆ ಅನ್ವಯಿಸುವುದಿಲ್ಲ

Water Analyst: Ramya Krishna K

ಪ್ರಾಧ್ಯಾಪಕರು ಮತ್ತು ಮುಖ್ಯಸ್ಥರು

As per ICAR Critical Limits Soil Parameters and it's Interpretation

Limits of Nutrients in Soil /ಮಣ್ಣಿನಲ್ಲಿನ ಪೋಷಕಾಂಶಗಳ ಮಿತಿಗಳು

Parameters				
pH	ರಸಸಾರ	<6.5 ಹಳಿ	6.5-7.5 ತಟಸ್ಥ	>8.5 ಕ್ಷಾರ
EC (dS/m)	ವಿದ್ಯುತ್ ವಾಹಕತ್ವ,ಡೆಸಿ,ಸೈಮನ್/ಮೀ	<0.8 ಸಹಜ ಸ್ಥಿತಿ	0.8-1.6	>1.6-2.5 ಹಾನಿಕಾರಕ

Parameters		Low/ಕಡಿಮೆ	Medium/ಮಧ್ಯಮ	High/ಹೆಚ್ಚು
OC (Organic Carbon %)	ಸಾವಯವ ಇಂಗಾಲ (%)	<0.50	0.5 - 0.75	>0.75
Nitrogen (Kg ha ⁻¹)	ದೊರೆಯುವ ಸಾರಜನಕ (N) ಕೆ.ಜಿ./ಹೆ.	<280	280-560	>560
Phosphorus (Kg ha ⁻¹)	ದೊರೆಯುವ ರಂಜಕ (P ₂ O ₅) ಕೆ.ಜಿ./ಹೆ.	<22.9	22.9-56.33	>56.33
Potassium (Kg ha ⁻¹)	ದೊರೆಯುವ ಪೊಟ್ಯಾಶ್ (K ₂ O) ಕೆ.ಜಿ./ಹೆ.	<141	141-336	>336
Calcium	ವಿನಿಮಯ ಕ್ಯಾಲ್ಸಿಯಂ (ಮಿ.ಈ/100 ಗ್ರಾಂ)	<1.5	1.5-3.0	>3.0
Magnesium	ವಿನಿಮಯ ಮೆಗ್ನೀಷಿಯಂ (ಮಿ.ಈ/100 ಗ್ರಾಂ)	<1.0	1.0-2.0	>2.0
Sulphur (ppm)/mg kg ⁻¹	ದೊರೆಯುವ ಗಂಧಕ(S) ಪಿ.ಪಿ.ಎಂ	<10	10-20	>20
Iron (ppm)/mg kg ⁻¹	ಕಬ್ಬಿಣ (Fe) ಪಿ.ಪಿ.ಎಂ	<2.50	2.50-4.50	>4.50
Manganese (ppm)/mg kg ⁻¹	ಮ್ಯಾಂಗನೀಸ್(Mn) ಪಿ.ಪಿ.ಎಂ	<1.00	1.00-2.00	>2.00
Copper (ppm)/mg kg ⁻¹	ತಾಮ್ರ(Cu) ಪಿ.ಪಿ.ಎಂ	<0.10	0.10-0.20	>0.20
Zinc (ppm)/mg kg ⁻¹	ಸತು (Zn) ಪಿ.ಪಿ.ಎಂ	<0.60	0.60-1.00	>1.00
Boron (ppm)/mg kg ⁻¹	ಬೋರಾನ್ (B) ಪಿ.ಪಿ.ಎಂ	<0.25	0.25-0.50	>0.50

Irrigation Water Quality Parameters/ ನೀರಾವರಿ ನೀರಿನ ಗುಣಮಟ್ಟ ನಿಯತಾಂಕ

Parameter	Low/ಕಡಿಮೆ	Medium/ಮಧ್ಯಮ	High/ಹೆಚ್ಚು	Very High/ಅತಿ ಹೆಚ್ಚು
pH/ರಸಸಾರ	<6.5	6.5-7.5	>7.5	-
EC(dS/M) ವಿದ್ಯುತ್ ವಾಹಕತ್ವ	<0.25	0.25-0.75	0.75-2.25	>2.25
Chlorides(Cl)(ppm)/ಕ್ಲೋರೈಡ್	<2-5	5-12	12-20	>20
SAR(Sodium adsorption ratio) ಸೋಡಿಯಂ ಹೊರಹೀರುವಿಕೆ ಅನುಪಾತ	<10	10-18	18-26	>26
RSC(Residual Sodium carbonate) ಉಳಿದ ಸೋಡಿಯಂ ಕಾರ್ಬೋನೇಟ್	<1.25	1.25-2.50	>2.50	-

Based on the soil parameters provided for Chilmattur Mandal, Sri Sathya Sai District, Andhra Pradesh, here is the soil health analysis, nutrient assessment, and recommendations for sustainable cultivation.

Soil Health Report Interpretation

1. pH (6.84): Neutral. This is ideal for most crops, ensuring maximum nutrient availability.
2. EC (0.15 dS/m): Low/Normal. The soil is not saline, which is excellent for plant root health.
3. OC (0.40%): Low. Organic Carbon is below the desirable range (<0.5%). Increasing this is critical for better water retention and nutrient efficiency.
4. N (188 kg/ha): Low. Nitrogen is deficient, which is common in red sandy loam soils of this region.
5. P (18.44 kg/ha): Low. Phosphorus is low and needs to be supplemented to support root development.
6. K (155 kg/ha): Medium. Potassium levels are adequate for many crops.
7. Micronutrients:
 - Fe (6.51 mg/kg): Sufficient (>4.0).
 - Mn (3.44 mg/kg): Sufficient (>1.0).
 - Cu (2.18 mg/kg): Sufficient (>0.2).
 - Zn (0.41 mg/kg): Deficient (<0.6). Zinc fertilization is recommended.

Soil Fertility Constraints

- **Low Organic Carbon:** Requires immediate addition of manure to improve soil structure and water retention.
- **Low Nitrogen and Phosphorus:** Requires balanced, site-specific fertilizer application.
- **Low Zinc:** Micronutrient deficiency needs to be addressed for higher yields.

Suitable Crops for Cultivation

The following crops are suitable for the region:

1. **Commercial/Cash Crops:** Groundnut, Cotton, Pigeon pea (Redgram).
2. **Millets & Pulses:** Finger millet (Ragi), Horse gram (due to tolerance to low fertility), Pearl millet.
3. **Horticulture:** Mango, Sapota, Cashew, Pomegranate.
4. **Vegetables:** Tomato, Onion, Green beans (when limited irrigation is available).

Recommendations for Management

- **Improve Organic Carbon:** Incorporate Farm Yard Manure (FYM), compost, or green manures (e.g., Sunnhemp) to raise the OC to over 0.5%.
- **Balanced Fertilization:** Apply N and P fertilizers based on soil test values. Increase Nitrogen application to meet the crop demand.
- **Correct Deficiencies:** Apply Zinc Sulfate (ZnSO₄) as a basal dose to correct Zinc deficiency.
- **In-situ Moisture Conservation:** Use mulching and adopt reduced tillage to prevent soil erosion and conserve moisture in the rainfed areas of Sri Sathya Sai district.

Water report

Based on the parameters provided, the water quality is generally suitable to marginally suitable for irrigation, depending on soil type and drainage. The most critical constraint is the moderate to high salinity (EC 1.99), which requires management to prevent salt accumulation in the root zone.

Water Quality Report Summary

1. pH: 7.01 (Neutral): Ideal range for irrigation water (6.5–8.4).
2. EC: 1.99 dS/m (High Salinity - C3): The water is in the range of 750-2250 S/cm (0.75-2.25 dS/m), classified as "Permissible" to "Marginal". It can be used for salt-tolerant crops on permeable soils but may damage salt-sensitive crops.
3. SAR: 0.96 (Low Sodium Hazard - S1): Very low sodium hazard. It will not cause soil structural degradation or infiltration issues.
4. RSC: Negative (Safe): Residual Sodium Carbonate is negative, meaning calcium and magnesium exceed bicarbonates, which is very good. This indicates no risk of creating an alkaline/sodic soil condition.
5. Chlorides: 10.40 meq/L (Moderate to Severe): Moderate to high levels (>4 meq/L). Potential for leaf burn on sensitive crops (like citrus, grapes) if used with sprinklers.

Suitability and Management Recommendations

1. **Soil Type:** Best suited for sandy loam or well-drained soils. Avoid use on heavy clay or poorly drained soils.
2. **Crop Selection:** Use for crops with moderate to high salt tolerance.
3. **Leaching Requirement:** Because the EC is high (1.99), it is necessary to apply extra water to leach salts below the root zone, particularly if there is no high rainfall.
4. **Irrigation Method:** Surface (furrow) or drip irrigation is preferred over sprinklers to avoid leaf tip burn from chloride